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<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 1400 1050" font-family="'Seg
<defs>
  <style>
    .pin-label { font-size: 10px; fill: #333; }
    .pin-num { font-size: 9px; fill: #888; font-weight: bold; }
    .component-title { font-size: 14px; fill: #fff; font-weight: bold; }
    .component-detail { font-size: 10px; fill: #e0e0e0; }
    .wire-i2c { stroke: #2196F3; stroke-width: 2.5; fill: none; }
    .wire-i2s { stroke: #9C27B0; stroke-width: 2.5; fill: none; }
    .wire-gpio { stroke: #4CAF50; stroke-width: 2.5; fill: none; }
    .wire-power { stroke: #F44336; stroke-width: 2.5; fill: none; }
    .wire-gnd { stroke: #333; stroke-width: 2.5; fill: none; }
    .wire-usb { stroke: #FF9800; stroke-width: 2.5; fill: none; }
    .legend-text { font-size: 11px; fill: #333; }
    .section-label { font-size: 12px; fill: #666; font-weight: bold; text-trans
    .note-text { font-size: 10px; fill: #555; }
    .addr-text { font-size: 9px; fill: #FFD54F; font-weight: bold; }
  </style>
  <marker id="arrowBlue" markerWidth="8" markerHeight="6" refX="8" refY="3" ori
    <polygon points="0 0, 8 3, 0 6" fill="#2196F3"/>
  </marker>
  <marker id="arrowPurple" markerWidth="8" markerHeight="6" refX="8" refY="3" o
    <polygon points="0 0, 8 3, 0 6" fill="#9C27B0"/>
  </marker>
  <marker id="arrowGreen" markerWidth="8" markerHeight="6" refX="8" refY="3" or
    <polygon points="0 0, 8 3, 0 6" fill="#4CAF50"/>
  </marker>
  <marker id="arrowRed" markerWidth="8" markerHeight="6" refX="8" refY="3" orie
    <polygon points="0 0, 8 3, 0 6" fill="#F44336"/>
  </marker>
  <marker id="arrowOrange" markerWidth="8" markerHeight="6" refX="8" refY="3" o
    <polygon points="0 0, 8 3, 0 6" fill="#FF9800"/>
  </marker>
  <marker id="arrowBlack" markerWidth="8" markerHeight="6" refX="8" refY="3" or
    <polygon points="0 0, 8 3, 0 6" fill="#333"/>
  </marker>
</defs>

<!-- Background -->
<rect width="1400" height="1050" fill="#FAFAFA" rx="8"/>

<!-- Title -->
<text x="700" y="35" text-anchor="middle" font-size="20" font-weight="bold" fil
<text x="700" y="55" text-anchor="middle" font-size="12" fill="#777">Raspberry

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<!-- ===== PI ZERO 2 W BOARD ===== -->
<g transform="translate(420, 90)">
  <!-- Board outline -->
  <rect x="0" y="0" width="300" height="560" rx="8" fill="#2E7D32" stroke="#1B5
  <rect x="10" y="10" width="280" height="540" rx="4" fill="#388E3C" opacity="0

  <!-- Board title -->
  <text x="150" y="35" text-anchor="middle" font-size="16" font-weight="bold" f
  <text x="150" y="52" text-anchor="middle" font-size="10" fill="#A5D6A7">BCM27

  <!-- USB OTG port -->
  <rect x="105" y="530" width="50" height="22" rx="3" fill="#555" stroke="#333"
  <text x="130" y="545" text-anchor="middle" font-size="8" fill="#fff">USB OTG<

  <!-- USB Power port -->
  <rect x="175" y="530" width="50" height="22" rx="3" fill="#555" stroke="#333"
  <text x="200" y="545" text-anchor="middle" font-size="8" fill="#fff">USB PWR<

  <!-- CSI camera connector (unused) -->
  <rect x="120" y="495" width="60" height="10" rx="2" fill="#444" stroke="#333"
  <text x="150" y="520" text-anchor="middle" font-size="7" fill="#81C784">CSI (

  <!-- GPIO Header - Left column (odd pins: 1,3,5,...,39) -->
  <!-- Pin 1: 3.3V -->
  <circle cx="55" cy="85" r="6" fill="#F44336" stroke="#C62828" stroke-width="1
  <text x="45" y="89" text-anchor="end" class="pin-label">3V3</text>
  <text x="55" y="79" text-anchor="middle" class="pin-num">1</text>

  <!-- Pin 3: GPIO2 (SDA) -->
  <circle cx="55" cy="110" r="6" fill="#2196F3" stroke="#1565C0" stroke-width="
  <text x="45" y="114" text-anchor="end" class="pin-label">SDA1</text>
  <text x="55" y="104" text-anchor="middle" class="pin-num">3</text>

  <!-- Pin 5: GPIO3 (SCL) -->
  <circle cx="55" cy="135" r="6" fill="#2196F3" stroke="#1565C0" stroke-width="
  <text x="45" y="139" text-anchor="end" class="pin-label">SCL1</text>
  <text x="55" y="129" text-anchor="middle" class="pin-num">5</text>

  <!-- Pin 7: GPIO4 -->
  <circle cx="55" cy="160" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
  <text x="45" y="164" text-anchor="end" class="pin-label">GPIO4</text>
  <text x="55" y="154" text-anchor="middle" class="pin-num">7</text>

  <!-- Pin 9: GND -->
  <circle cx="55" cy="185" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
  <text x="45" y="189" text-anchor="end" class="pin-label">GND</text>

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<text x="55" y="179" text-anchor="middle" class="pin-num">9</text>

<!-- Pin 11: GPIO17 -->
<circle cx="55" cy="210" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="45" y="214" text-anchor="end" class="pin-label">GPIO17</text>
<text x="55" y="204" text-anchor="middle" class="pin-num">11</text>

<!-- Pin 13: GPIO27 -->
<circle cx="55" cy="235" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="45" y="239" text-anchor="end" class="pin-label">GPIO27</text>
<text x="55" y="229" text-anchor="middle" class="pin-num">13</text>

<!-- Pin 15: GPIO22 -->
<circle cx="55" cy="260" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="45" y="264" text-anchor="end" class="pin-label">GPIO22</text>
<text x="55" y="254" text-anchor="middle" class="pin-num">15</text>

<!-- Pin 17: 3.3V -->
<circle cx="55" cy="285" r="6" fill="#F44336" stroke="#C62828" stroke-width="
<text x="45" y="289" text-anchor="end" class="pin-label">3V3</text>
<text x="55" y="279" text-anchor="middle" class="pin-num">17</text>

<!-- Pin 19: GPIO10 (MOSI) -->
<circle cx="55" cy="310" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="314" text-anchor="end" class="pin-label">MOSI</text>
<text x="55" y="304" text-anchor="middle" class="pin-num">19</text>

<!-- Pin 21: GPIO9 (MISO) -->
<circle cx="55" cy="335" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="339" text-anchor="end" class="pin-label">MISO</text>
<text x="55" y="329" text-anchor="middle" class="pin-num">21</text>

<!-- Pin 23: GPIO11 (SCLK) -->
<circle cx="55" cy="360" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="364" text-anchor="end" class="pin-label">SCLK</text>
<text x="55" y="354" text-anchor="middle" class="pin-num">23</text>

<!-- Pin 25: GND -->
<circle cx="55" cy="385" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="45" y="389" text-anchor="end" class="pin-label">GND</text>
<text x="55" y="379" text-anchor="middle" class="pin-num">25</text>

<!-- Pin 27: GPIO0 (ID_SD) -->
<circle cx="55" cy="410" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="414" text-anchor="end" class="pin-label">ID_SD</text>
<text x="55" y="404" text-anchor="middle" class="pin-num">27</text>

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<!-- Pin 29: GPIO5 -->
<circle cx="55" cy="435" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="45" y="439" text-anchor="end" class="pin-label">GPIO5</text>
<text x="55" y="429" text-anchor="middle" class="pin-num">29</text>

<!-- Pin 31: GPIO6 -->
<circle cx="55" cy="460" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="464" text-anchor="end" class="pin-label">GPIO6</text>
<text x="55" y="454" text-anchor="middle" class="pin-num">31</text>

<!-- Pin 33: GPIO13 -->
<circle cx="55" cy="485" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="489" text-anchor="end" class="pin-label">GPIO13</text>
<text x="55" y="479" text-anchor="middle" class="pin-num">33</text>

<!-- GPIO Header - Right column (even pins: 2,4,6,...,40) -->
<!-- Pin 2: 5V -->
<circle cx="85" cy="85" r="6" fill="#D32F2F" stroke="#B71C1C" stroke-width="1
<text x="95" y="89" text-anchor="start" class="pin-label">5V</text>
<text x="85" y="79" text-anchor="middle" class="pin-num">2</text>

<!-- Pin 4: 5V -->
<circle cx="85" cy="110" r="6" fill="#D32F2F" stroke="#B71C1C" stroke-width="
<text x="95" y="114" text-anchor="start" class="pin-label">5V</text>
<text x="85" y="104" text-anchor="middle" class="pin-num">4</text>

<!-- Pin 6: GND -->
<circle cx="85" cy="135" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="95" y="139" text-anchor="start" class="pin-label">GND</text>
<text x="85" y="129" text-anchor="middle" class="pin-num">6</text>

<!-- Pin 8: GPIO14 (TXD) -->
<circle cx="85" cy="160" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="164" text-anchor="start" class="pin-label">TXD</text>
<text x="85" y="154" text-anchor="middle" class="pin-num">8</text>

<!-- Pin 10: GPIO15 (RXD) -->
<circle cx="85" cy="185" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="189" text-anchor="start" class="pin-label">RXD</text>
<text x="85" y="179" text-anchor="middle" class="pin-num">10</text>

<!-- Pin 12: GPIO18 (PCM_CLK / I2S BCLK) -->
<circle cx="85" cy="210" r="6" fill="#9C27B0" stroke="#6A1B9A" stroke-width="
<text x="95" y="214" text-anchor="start" class="pin-label">GPIO18</text>
<text x="85" y="204" text-anchor="middle" class="pin-num">12</text>

<!-- Pin 14: GND -->

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<circle cx="85" cy="235" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="95" y="239" text-anchor="start" class="pin-label">GND</text>
<text x="85" y="229" text-anchor="middle" class="pin-num">14</text>

<!-- Pin 16: GPIO23 -->
<circle cx="85" cy="260" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="95" y="264" text-anchor="start" class="pin-label">GPIO23</text>
<text x="85" y="254" text-anchor="middle" class="pin-num">16</text>

<!-- Pin 18: GPIO24 -->
<circle cx="85" cy="285" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="95" y="289" text-anchor="start" class="pin-label">GPIO24</text>
<text x="85" y="279" text-anchor="middle" class="pin-num">18</text>

<!-- Pin 20: GND -->
<circle cx="85" cy="310" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="95" y="314" text-anchor="start" class="pin-label">GND</text>
<text x="85" y="304" text-anchor="middle" class="pin-num">20</text>

<!-- Pin 22: GPIO25 -->
<circle cx="85" cy="335" r="6" fill="#4CAF50" stroke="#2E7D32" stroke-width="
<text x="95" y="339" text-anchor="start" class="pin-label">GPIO25</text>
<text x="85" y="329" text-anchor="middle" class="pin-num">22</text>

<!-- Pin 24: GPIO8 (CE0) -->
<circle cx="85" cy="360" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="364" text-anchor="start" class="pin-label">CE0</text>
<text x="85" y="354" text-anchor="middle" class="pin-num">24</text>

<!-- Pin 26: GPIO7 (CE1) -->
<circle cx="85" cy="385" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="389" text-anchor="start" class="pin-label">CE1</text>
<text x="85" y="379" text-anchor="middle" class="pin-num">26</text>

<!-- Pin 28: GPIO1 (ID_SC) -->
<circle cx="85" cy="410" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="414" text-anchor="start" class="pin-label">ID_SC</text>
<text x="85" y="404" text-anchor="middle" class="pin-num">28</text>

<!-- Pin 30: GND -->
<circle cx="85" cy="435" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="95" y="439" text-anchor="start" class="pin-label">GND</text>
<text x="85" y="429" text-anchor="middle" class="pin-num">30</text>

<!-- Pin 32: GPIO12 -->
<circle cx="85" cy="460" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="464" text-anchor="start" class="pin-label">GPIO12</text>

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<text x="85" y="454" text-anchor="middle" class="pin-num">32</text>

<!-- Pin 34: GND -->
<circle cx="85" cy="485" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="95" y="489" text-anchor="start" class="pin-label">GND</text>
<text x="85" y="479" text-anchor="middle" class="pin-num">34</text>

<!-- Pin 35: GPIO19 (I2S LRCLK) -->
<circle cx="55" cy="510" r="6" fill="#9C27B0" stroke="#6A1B9A" stroke-width="
<text x="45" y="504" text-anchor="end" class="pin-num">35</text>
<text x="45" y="514" text-anchor="end" class="pin-label">GPIO19</text>

<!-- Pin 36: GPIO16 -->
<circle cx="85" cy="510" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="514" text-anchor="start" class="pin-label">GPIO16</text>
<text x="85" y="504" text-anchor="middle" class="pin-num">36</text>

<!-- Pin 37: GPIO26 -->
<circle cx="55" cy="535" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="45" y="529" text-anchor="end" class="pin-num">37</text>

<!-- Pin 38: GPIO20 -->
<circle cx="85" cy="535" r="6" fill="#757575" stroke="#424242" stroke-width="
<text x="95" y="539" text-anchor="start" class="pin-label">GPIO20</text>
<text x="85" y="529" text-anchor="middle" class="pin-num">38</text>

<!-- Pin 39: GND -->
<circle cx="55" cy="555" r="6" fill="#333" stroke="#111" stroke-width="1.5"/>
<text x="45" y="549" text-anchor="end" class="pin-num">39</text>

<!-- Pin 40: GPIO21 (I2S DOUT) -->
<circle cx="85" cy="555" r="6" fill="#9C27B0" stroke="#6A1B9A" stroke-width="
<text x="95" y="559" text-anchor="start" class="pin-label">GPIO21</text>
<text x="85" y="549" text-anchor="middle" class="pin-num">40</text>

<!-- Highlight used pins with rings -->
<!-- SDA -->
<circle cx="55" cy="110" r="9" fill="none" stroke="#2196F3" stroke-width="1"
<!-- SCL -->
<circle cx="55" cy="135" r="9" fill="none" stroke="#2196F3" stroke-width="1"
<!-- GPIO4 - Button -->
<circle cx="55" cy="160" r="9" fill="none" stroke="#4CAF50" stroke-width="1"
<!-- GPIO17 - Encoder CLK -->
<circle cx="55" cy="210" r="9" fill="none" stroke="#4CAF50" stroke-width="1"
<!-- GPIO27 - Encoder DT -->
<circle cx="55" cy="235" r="9" fill="none" stroke="#4CAF50" stroke-width="1"
<!-- GPIO22 - Encoder SW -->

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<circle cx="55" cy="260" r="9" fill="none" stroke="#4CAF50" stroke-width="1"
<!-- GPIO5 - Motor -->
<circle cx="55" cy="435" r="9" fill="none" stroke="#4CAF50" stroke-width="1"
<!-- GPIO18 - BCLK -->
<circle cx="85" cy="210" r="9" fill="none" stroke="#9C27B0" stroke-width="1"
<!-- GPIO19 - LRCLK -->
<circle cx="55" cy="510" r="9" fill="none" stroke="#9C27B0" stroke-width="1"
<!-- GPIO21 - DIN -->
<circle cx="85" cy="555" r="9" fill="none" stroke="#9C27B0" stroke-width="1"
</g>

<!-- ===== COMPONENT BOXES ===== -->

<!-- I2C BUS LABEL -->
<text x="85" y="110" text-anchor="middle" class="section-label" fill="#2196F3">

<!-- MPU6050 -->
<g transform="translate(30, 120)">
  <rect x="0" y="0" width="180" height="80" rx="6" fill="#1565C0" stroke="#0D47
  <text x="90" y="22" text-anchor="middle" class="component-title">MPU6050 GY-5
  <text x="90" y="38" text-anchor="middle" class="component-detail">IMU - Postu
  <text x="90" y="52" text-anchor="middle" class="addr-text">I2C 0x68</text>
  <text x="90" y="68" text-anchor="middle" class="component-detail">VCC · GND ·
</g>

<!-- MAX30102 -->
<g transform="translate(30, 220)">
  <rect x="0" y="0" width="180" height="80" rx="6" fill="#1565C0" stroke="#0D47
  <text x="90" y="22" text-anchor="middle" class="component-title">MAX30102</te
  <text x="90" y="38" text-anchor="middle" class="component-detail">HRV - Clip
  <text x="90" y="52" text-anchor="middle" class="addr-text">I2C 0x57</text>
  <text x="90" y="68" text-anchor="middle" class="component-detail">VCC · GND ·
</g>

<!-- SSD1306 OLED -->
<g transform="translate(30, 320)">
  <rect x="0" y="0" width="180" height="80" rx="6" fill="#1565C0" stroke="#0D47
  <text x="90" y="22" text-anchor="middle" class="component-title">SSD1306 / SH
  <text x="90" y="38" text-anchor="middle" class="component-detail">1.3" OLED -
  <text x="90" y="52" text-anchor="middle" class="addr-text">I2C 0x3C</text>
  <text x="90" y="68" text-anchor="middle" class="component-detail">VCC · GND ·
</g>

<!-- I2C wires from components to Pi -->
<!-- MPU6050 SDA -->
<path d="M 210,155 L 320,155 Q 340,155 340,170 L 340,200 L 475,200" class="wire
<!-- MPU6050 SCL -->

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<path d="M 210,165 L 310,165 Q 330,165 330,180 L 330,225 L 475,225" class="wire
<!-- MAX30102 SDA -->
<path d="M 210,255 L 300,255 Q 320,255 320,240 L 320,200 L 475,200" class="wire
<!-- MAX30102 SCL -->
<path d="M 210,265 L 290,265 Q 310,265 310,250 L 310,225 L 475,225" class="wire
<!-- OLED SDA -->
<path d="M 210,355 L 280,355 Q 300,355 300,340 L 300,200 L 475,200" class="wire
<!-- OLED SCL -->
<path d="M 210,365 L 270,365 Q 290,365 290,350 L 290,225 L 475,225" class="wire

<!-- I2C bus labels on wires -->
<text x="400" y="195" text-anchor="middle" font-size="9" fill="#1565C0" font-we
<text x="400" y="220" text-anchor="middle" font-size="9" fill="#1565C0" font-we

<!-- 3.3V power lines to I2C devices -->
<path d="M 210,140 L 250,140 Q 260,140 260,130 L 260,95 L 380,95 L 380,175 L 47
<text x="350" y="88" font-size="8" fill="#F44336">3V3 (Pin 1)</text>

<!-- ===== I2S / AUDIO (RIGHT SIDE) ===== -->
<text x="1000" y="110" text-anchor="middle" class="section-label" fill="#9C27B0

<!-- MAX98357A -->
<g transform="translate(880, 120)">
  <rect x="0" y="0" width="200" height="100" rx="6" fill="#6A1B9A" stroke="#4A1
  <text x="100" y="22" text-anchor="middle" class="component-title">MAX98357A</
  <text x="100" y="38" text-anchor="middle" class="component-detail">I2S Amplif
  <text x="100" y="58" text-anchor="middle" class="component-detail">VIN: 5V ·
  <text x="100" y="78" text-anchor="middle" class="component-detail">GAIN: tied
  <text x="100" y="92" text-anchor="middle" class="component-detail">SD: tied t
</g>

<!-- Speaker -->
<g transform="translate(920, 240)">
  <rect x="0" y="0" width="140" height="60" rx="6" fill="#4A148C" stroke="#311B
  <text x="70" y="22" text-anchor="middle" class="component-title">Speaker</tex
  <text x="70" y="40" text-anchor="middle" class="component-detail">28-32mm · 8
  <text x="70" y="54" text-anchor="middle" class="component-detail">Bare wire f
</g>

<!-- I2S wires from Pi to MAX98357A -->
<!-- BCLK: GPIO18 (Pin 12) -->
<path d="M 505,300 L 760,300 L 760,155 L 880,155" class="wire-i2s" marker-end="
<text x="780" y="150" font-size="9" fill="#9C27B0" font-weight="bold">BCLK</tex
<!-- LRCLK: GPIO19 (Pin 35) -->
<path d="M 475,600 L 400,600 Q 380,600 380,580 L 380,560 Q 380,540 400,540 L 75
<text x="780" y="166" font-size="9" fill="#9C27B0" font-weight="bold">LRCLK</te
<!-- DIN: GPIO21 (Pin 40) -->

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<path d="M 505,645 L 780,645 Q 800,645 800,625 L 800,190 L 880,190" class="wire"
<text x="812" y="186" font-size="9" fill="#9C27B0" font-weight="bold">DIN</text>

<!-- 5V to amp -->
<path d="M 505,175 L 530,175 Q 550,175 550,160 L 550,140 L 880,140" class="wire"
<text x="700" y="135" font-size="8" fill="#D32F2F">5V (Pin 2)</text>

<!-- Speaker wire from amp -->
<path d="M 980,220 L 980,240" class="wire-i2s" stroke="#E1BEE7" marker-end="url

<!-- ===== GPIO DEVICES (RIGHT SIDE) ===== -->
<text x="1030" y="340" text-anchor="middle" class="section-label" fill="#4CAF50

<!-- Tactile Button -->
<g transform="translate(880, 355)">
  <rect x="0" y="0" width="200" height="75" rx="6" fill="#2E7D32" stroke="#1B5E
  <text x="100" y="22" text-anchor="middle" class="component-title">Tactile But
  <text x="100" y="38" text-anchor="middle" class="component-detail">Top-left ·
  <text x="100" y="55" text-anchor="middle" class="component-detail">GPIO4 (Pin
  <text x="100" y="68" text-anchor="middle" class="component-detail">Internal p
</g>

<!-- Button wire -->
<path d="M 475,250 L 420,250 Q 400,250 400,270 L 400,390 L 880,390" class="wire"
<text x="640" y="384" font-size="9" fill="#4CAF50" font-weight="bold">GPIO4 (Pi

<!-- EC11 Rotary Encoder -->
<g transform="translate(880, 450)">
  <rect x="0" y="0" width="200" height="95" rx="6" fill="#2E7D32" stroke="#1B5E
  <text x="100" y="22" text-anchor="middle" class="component-title">EC11 Encode
  <text x="100" y="38" text-anchor="middle" class="component-detail">Top-right
  <text x="100" y="55" text-anchor="middle" class="component-detail">CLK → GPIO
  <text x="100" y="68" text-anchor="middle" class="component-detail">DT → GPIO
  <text x="100" y="81" text-anchor="middle" class="component-detail">SW → GPIO
</g>

<!-- Encoder wires -->
<!-- CLK: GPIO17 -->
<path d="M 475,300 L 430,300 Q 410,300 410,320 L 410,475 L 880,475" class="wire"
<!-- DT: GPIO27 -->
<path d="M 475,325 L 440,325 Q 420,325 420,345 L 420,490 L 880,490" class="wire"
<!-- SW: GPIO22 -->
<path d="M 475,350 L 450,350 Q 430,350 430,370 L 430,505 L 880,505" class="wire

<!-- 10nF debounce note -->
<g transform="translate(880, 550)">
  <rect x="0" y="0" width="200" height="32" rx="4" fill="#E8F5E9" stroke="#A5D6

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    <text x="100" y="20" text-anchor="middle" class="note-text" fill="#2E7D32">10
</g>

<!-- ===== VIBRATION MOTOR (BOTTOM LEFT) ===== -->
<g transform="translate(30, 470)">
    <rect x="0" y="0" width="200" height="120" rx="6" fill="#E65100" stroke="#BF3
    <text x="100" y="22" text-anchor="middle" class="component-title">Coin Vibrat
    <text x="100" y="40" text-anchor="middle" class="component-detail">Haptic nud
    <text x="100" y="60" text-anchor="middle" class="component-detail" fill="#FFE
    <text x="100" y="76" text-anchor="middle" class="component-detail" fill="#FFE
    <text x="100" y="92" text-anchor="middle" class="component-detail" fill="#FFE
    <text x="100" y="108" text-anchor="middle" class="component-detail" fill="#FF
</g>

<!-- Motor wire: GPIO5 (Pin 29) -->
<path d="M 475,525 L 350,525 Q 330,525 330,530 L 330,530 L 230,530" class="wire
<text x="340" y="520" font-size="9" fill="#4CAF50" font-weight="bold">GPIO5 (Pi

<!-- ===== USB CAMERA (BOTTOM) ===== -->
<g transform="translate(30, 620)">
    <rect x="0" y="0" width="200" height="80" rx="6" fill="#E65100" stroke="#BF36
    <text x="100" y="22" text-anchor="middle" class="component-title">OV9712 Came
    <text x="100" y="38" text-anchor="middle" class="component-detail">720p USB U
    <text x="100" y="56" text-anchor="middle" class="component-detail" fill="#FFE
    <text x="100" y="72" text-anchor="middle" class="component-detail" fill="#FFE
</g>

<!-- USB wire to Pi OTG -->
<path d="M 230,660 L 360,660 Q 380,660 380,680 L 380,700 Q 380,710 400,710 L 52
<text x="480" y="705" font-size="9" fill="#FF9800" font-weight="bold">USB OTG</

<!-- ===== POWER (BOTTOM RIGHT) ===== -->
<g transform="translate(880, 620)">
    <rect x="0" y="0" width="200" height="80" rx="6" fill="#B71C1C" stroke="#880E
    <text x="100" y="22" text-anchor="middle" class="component-title">Power Bank<
    <text x="100" y="38" text-anchor="middle" class="component-detail">10,000 mAh
    <text x="100" y="56" text-anchor="middle" class="component-detail" fill="#FFC
    <text x="100" y="72" text-anchor="middle" class="component-detail" fill="#FFC
</g>

<!-- Power wire -->
<path d="M 880,660 L 620,660 Q 600,660 600,650 L 600,640" class="wire-power" ma
<text x="740" y="654" font-size="9" fill="#D32F2F" font-weight="bold">USB-C + X

<!-- ===== LEGEND ===== -->
<g transform="translate(30, 760)">
    <rect x="0" y="0" width="1340" height="70" rx="6" fill="#fff" stroke="#ddd" s

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<text x="20" y="20" font-size="12" font-weight="bold" fill="#333">Legend</tex

<line x1="20" y1="40" x2="60" y2="40" class="wire-i2c"/>
<text x="68" y="44" class="legend-text">I2C (SDA/SCL shared bus)</text>

<line x1="240" y1="40" x2="280" y2="40" class="wire-i2s"/>
<text x="288" y="44" class="legend-text">I2S (BCLK/LRCLK/DIN)</text>

<line x1="470" y1="40" x2="510" y2="40" class="wire-gpio"/>
<text x="518" y="44" class="legend-text">GPIO (digital I/O)</text>

<line x1="650" y1="40" x2="690" y2="40" class="wire-power"/>
<text x="698" y="44" class="legend-text">Power (3V3 / 5V)</text>

<line x1="820" y1="40" x2="860" y2="40" class="wire-gnd"/>
<text x="868" y="44" class="legend-text">Ground</text>

<line x1="960" y1="40" x2="1000" y2="40" class="wire-usb"/>
<text x="1008" y="44" class="legend-text">USB</text>

<circle cx="1120" cy="40" r="5" fill="#757575"/>
<text x="1132" y="44" class="legend-text">Unused pin</text>
</g>

<!-- ===== PIN ASSIGNMENT TABLE ===== -->
<g transform="translate(30, 850)">
  <rect x="0" y="0" width="1340" height="180" rx="6" fill="#fff" stroke="#ddd"
  <text x="20" y="24" font-size="13" font-weight="bold" fill="#333">Complete Pi

  <!-- Header -->
  <text x="20" y="50" font-size="10" font-weight="bold" fill="#555">Pin</text>
  <text x="70" y="50" font-size="10" font-weight="bold" fill="#555">GPIO</text>
  <text x="140" y="50" font-size="10" font-weight="bold" fill="#555">Function</
  <text x="320" y="50" font-size="10" font-weight="bold" fill="#555">Component<

  <text x="500" y="50" font-size="10" font-weight="bold" fill="#555">Pin</text>
  <text x="550" y="50" font-size="10" font-weight="bold" fill="#555">GPIO</text>
  <text x="620" y="50" font-size="10" font-weight="bold" fill="#555">Function</
  <text x="800" y="50" font-size="10" font-weight="bold" fill="#555">Component<

  <line x1="20" y1="56" x2="470" y2="56" stroke="#eee" stroke-width="1"/>
  <line x1="500" y1="56" x2="970" y2="56" stroke="#eee" stroke-width="1"/>

  <!-- Left column data -->
  <text x="20" y="72" font-size="10" fill="#F44336" font-weight="bold">1</text>
  <text x="70" y="72" font-size="10" fill="#333">—</text>
  <text x="140" y="72" font-size="10" fill="#333">3.3V Power</text>

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<text x="320" y="72" font-size="10" fill="#333">MPU6050, MAX30102, OLED, Moto

<text x="20" y="88" font-size="10" fill="#2196F3" font-weight="bold">3</text>
<text x="70" y="88" font-size="10" fill="#333">GPIO2</text>
<text x="140" y="88" font-size="10" fill="#333">I2C SDA</text>
<text x="320" y="88" font-size="10" fill="#333">MPU6050 + MAX30102 + OLED</te

<text x="20" y="104" font-size="10" fill="#2196F3" font-weight="bold">5</text>
<text x="70" y="104" font-size="10" fill="#333">GPIO3</text>
<text x="140" y="104" font-size="10" fill="#333">I2C SCL</text>
<text x="320" y="104" font-size="10" fill="#333">MPU6050 + MAX30102 + OLED</t

<text x="20" y="120" font-size="10" fill="#4CAF50" font-weight="bold">7</text>
<text x="70" y="120" font-size="10" fill="#333">GPIO4</text>
<text x="140" y="120" font-size="10" fill="#333">Button (interrupt)</text>
<text x="320" y="120" font-size="10" fill="#333">Tactile button → GND</text>

<text x="20" y="136" font-size="10" fill="#4CAF50" font-weight="bold">11</tex>
<text x="70" y="136" font-size="10" fill="#333">GPIO17</text>
<text x="140" y="136" font-size="10" fill="#333">Encoder CLK</text>
<text x="320" y="136" font-size="10" fill="#333">EC11 + 10nF to GND</text>

<text x="20" y="152" font-size="10" fill="#4CAF50" font-weight="bold">13</tex>
<text x="70" y="152" font-size="10" fill="#333">GPIO27</text>
<text x="140" y="152" font-size="10" fill="#333">Encoder DT</text>
<text x="320" y="152" font-size="10" fill="#333">EC11 + 10nF to GND</text>

<text x="20" y="168" font-size="10" fill="#4CAF50" font-weight="bold">15</tex>
<text x="70" y="168" font-size="10" fill="#333">GPIO22</text>
<text x="140" y="168" font-size="10" fill="#333">Encoder SW (click)</text>
<text x="320" y="168" font-size="10" fill="#333">EC11 push-button</text>

<!-- Right column data -->
<text x="500" y="72" font-size="10" fill="#D32F2F" font-weight="bold">2</text>
<text x="550" y="72" font-size="10" fill="#333">—</text>
<text x="620" y="72" font-size="10" fill="#333">5V Power</text>
<text x="800" y="72" font-size="10" fill="#333">MAX98357A (VIN)</text>

<text x="500" y="88" font-size="10" fill="#9C27B0" font-weight="bold">12</tex>
<text x="550" y="88" font-size="10" fill="#333">GPIO18</text>
<text x="620" y="88" font-size="10" fill="#333">I2S BCLK</text>
<text x="800" y="88" font-size="10" fill="#333">MAX98357A</text>

<text x="500" y="104" font-size="10" fill="#9C27B0" font-weight="bold">35</te>
<text x="550" y="104" font-size="10" fill="#333">GPIO19</text>
<text x="620" y="104" font-size="10" fill="#333">I2S LRCLK</text>
<text x="800" y="104" font-size="10" fill="#333">MAX98357A</text>

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<text x="500" y="120" font-size="10" fill="#9C27B0" font-weight="bold">40</te
<text x="550" y="120" font-size="10" fill="#333">GPIO21</text>
<text x="620" y="120" font-size="10" fill="#333">I2S DIN</text>
<text x="800" y="120" font-size="10" fill="#333">MAX98357A</text>

<text x="500" y="136" font-size="10" fill="#4CAF50" font-weight="bold">29</te
<text x="550" y="136" font-size="10" fill="#333">GPIO5</text>
<text x="620" y="136" font-size="10" fill="#333">Motor drive</text>
<text x="800" y="136" font-size="10" fill="#333">→ 1kΩ → 2N2222 base</text>

<text x="500" y="152" font-size="10" fill="#333" font-weight="bold">6,9,14...
<text x="550" y="152" font-size="10" fill="#333">—</text>
<text x="620" y="152" font-size="10" fill="#333">GND</text>
<text x="800" y="152" font-size="10" fill="#333">All components share ground<

<text x="500" y="168" font-size="10" fill="#FF9800" font-weight="bold">USB</t
<text x="550" y="168" font-size="10" fill="#333">OTG</text>
<text x="620" y="168" font-size="10" fill="#333">USB device port</text>
<text x="800" y="168" font-size="10" fill="#333">OV9712 camera (flat ribbon)<

<!-- Unused pins note -->
<text x="1050" y="72" font-size="10" fill="#999" font-style="italic">Unused G
<text x="1050" y="88" font-size="10" fill="#999" font-style="italic">(SPI, UA
<text x="1050" y="104" font-size="10" fill="#999" font-style="italic">Availab
<text x="1050" y="130" font-size="10" fill="#999" font-style="italic">Total p
<text x="1050" y="146" font-size="10" fill="#999" font-style="italic">+ 2 pow
</g>
</svg>

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